Stacking cell evaluation & gate audit

baseline (v1) at n=50 · commit 69d6830 · mujoco 3.11.0 · tip probability 0.2

SUCCESS RATE (N=50)

10%

5 of 50 episodes succeeded. The honest range around that number (95% confidence interval, Wilson): **4%–21%**

In other words: 50 episodes can't tell a 4% policy from a 21% one. A later 1,000-episode run pinned this same policy at 5.3% — see the gate audit below.

Production gate (≥90% at n=50): **does not pass**

HOW HIGH THE STACKS GOT

5·13·26·6

5 episodes ended with all three cylinders stacked, 13 with two, 26 with one, 6 with none. Partial stacks are counted here, on their own — they are never mixed into the headline success rate.

COST

1s

seconds of real time per episode — the full 50-episode run took 48 seconds on a laptop

Why episodes failed, ranked (45 failures)

tipped part skipped — 28

a cylinder spawned lying down; the policy has no strategy for tipped parts and skipped it

stack collapse — 6

placements destroyed the stack — one cylinder left standing at best

grasp miss — 5

the gripper closed but captured nothing (approach missed the part)

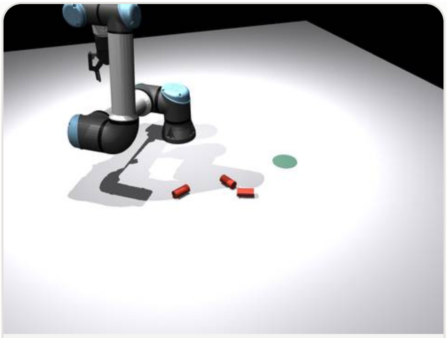
top placement failed — 5

two stacked cleanly; the third placement failed or knocked itself off

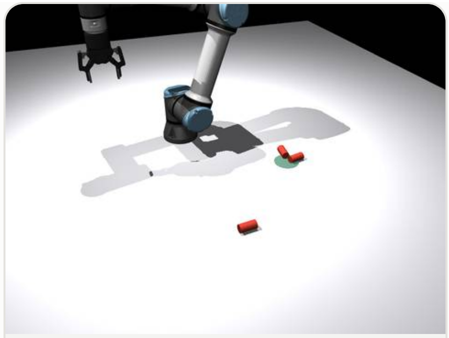
base misplaced — 1

everything was lifted and placed, but no cylinder ended within tolerance of the target

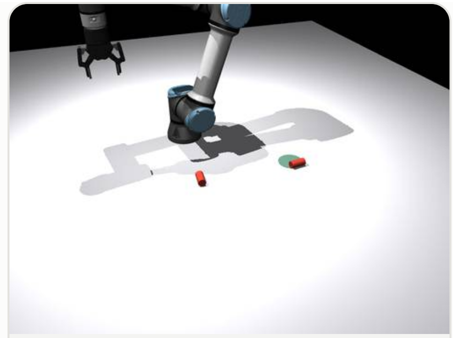
The worst episodes, first — each one replayable with a single command



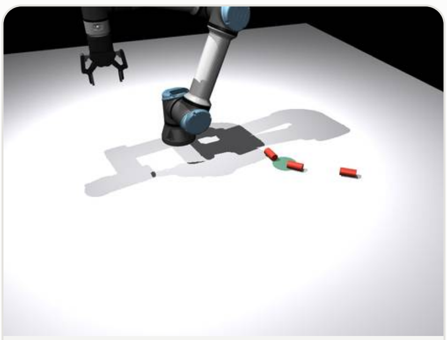
seed 2 — stacked 0 of 3 — tipped part skipped
watch it happen: `python render_video.py 2 --policy v1`



seed 6 — stacked 0 of 3 — grasp miss
watch it happen: `python render_video.py 6 --policy v1`



seed 9 — stacked 0 of 3 — tipped part skipped
watch it happen: `python render_video.py 9 --policy v1`



seed 41 — stacked 0 of 3 — base misplaced
watch it happen: `python render_video.py 41 --policy v1`

The gate audit

Production rule under test: ship if ≥45 of 50 episodes succeed.

What 50 episodes said vs. what 1,000 revealed

POLICY	SMALL-SAMPLE MEASURED	TRUE RATE (N=1000)	95% CI
v1	10% (n=50)	**5.3%**	4.1%–6.9%
v2	10% (n=50)	**6.6%**	5.2%–8.3%
mild (2mm noise)	6% (n=100)	**5.7%**	4.4%–7.3%
heavy (5mm noise)	2% (n=100)	**1.7%**	1.1%–2.7%

> The gate-sized sample overestimated v1 **2×** (10% vs 5.3%). v2 trends better than v1 (6.6% vs 5.3%) but **even n=1000 does not resolve them** — their CIs overlap, as do the mild-noise variant's. Three mechanically different policies whose success counts never separate: the gate is mechanism-blind; the failure taxonomy above is what tells these policies apart.

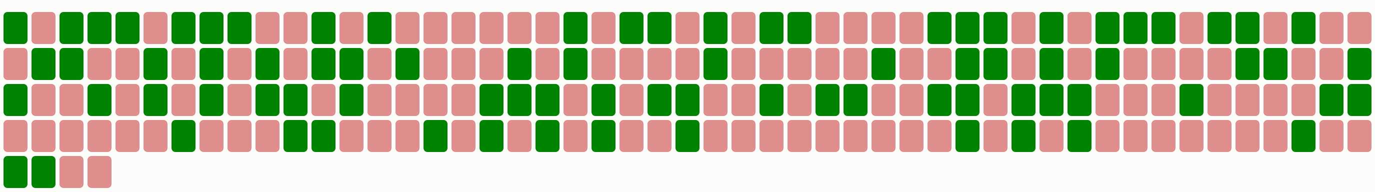
Two policy variants, side by side — every gate verdict

v1 — true rate 5.3% — 20 disjoint n=50 gates — 0/20 gates passed

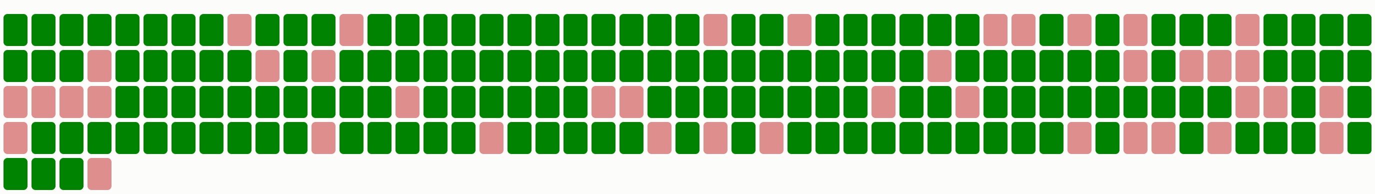
v2 — true rate 6.6% — 20 disjoint n=50 gates — 0/20 gates passed

> Far from the threshold the gate is reliable: two below-spec policies, 40 gates, zero false passes. The gate's errors live near the threshold — probed below with calibrated reference policies (an instrument we declare openly: seeded dial-a-rate policies whose stacks are judged by the same judge as everything else).

Watch the gate flip — calibrated policies near the threshold



calibrated reference policy, true rate 88% — measured pass rate 41.0% (exact theory: 43.5%) — 82/200 gates passed



calibrated reference policy, true rate 92% — measured pass rate 79.5% (exact theory: 79.2%) — 159/200 gates passed

> **A below-spec 88% policy ships ~4 times in 10. A policy at exactly 90% spec is rejected ~4 times in 10.** Each square is a full 50-episode gate on its own seeds, judged episode by episode.

Measured vs. theory, all levels

TRUE RATE	PIPELINE (200 GATES)	COIN-FLIP (20K GATES)	EXACT THEORY	DISPERSION RATIO
80%	6.0%	4.9%	4.8%	1.04
85%	25.0%	21.4%	21.9%	0.98
88%	41.0%	43.4%	43.5%	1.08

90%	64.5%	62.2%	61.6%	1.12
92%	79.5%	79.0%	79.2%	0.94
95%	96.0%	96.3%	96.2%	1.03

> Instrument calibration: dialed 88%, pipeline measured **88.4%** over n=3000 judged episodes. Scope, stated plainly: the reference policy draws each outcome from a seeded rate and arranges the scene for the real judge to rule — so this sweep validates the **measurement chain** (judge, gate arithmetic, seeding) against a known truth; by construction it cannot disagree with the coin-flip model unless the judge misrules an arranged scene. The gate's error rates therefore rest on the binomial model plus its independence assumption — which is checked on the **real** policies (dispersion ratios ≈1: v1 0.93, v2 1.06, mild 0.99, heavy 1.17), not assumed.

Reproduce

Every number regenerable from a fresh clone — commands in the README. Declared blind spot: grasping uses a kinematic attach within 20 mm; grasp-slip and drop-in-transit failures cannot occur in this evaluation.